

RODRIGO FARIA

AI Engineer | Software Engineer | .NET | C# | LLMs

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PROFESSIONAL SUMMARY

AI and Software Engineer with hands-on experience building backend systems and AI-powered applications using Python, C#, .NET, REST APIs and microservices. Professional experience developing LLM-based systems with Retrieval-Augmented Generation (RAG), vector search, semantic retrieval, AI automation and tool calling. Experienced in designing backend services, integrating external APIs and services, working with databases, and applying clean architecture and API-first design. Strong practical experience with LLM applications and AI agents through both professional work and personal projects, including RAG pipelines, agent orchestration, tool calling and intelligent workflows.

Computer Engineering graduate with a strong interest in building production-oriented AI products and backend systems, with particular interest in the intersection of .NET development and AI-powered automation.

EXPERIENCE

AI Engineer Intern

Feb. 2026 – Jun. 2026 *It's Possible Tech*

Portugal

- Designed and developed AI-powered backend services using Python and FastAPI, integrating Large Language Models, Retrieval-Augmented Generation (RAG), vector search and external APIs.
- Built end-to-end document processing pipelines including parsing, chunking, embedding generation, indexing and semantic retrieval over domain-specific documentation.
- Designed backend services within a microservices architecture, applying clean architecture, API-first design and separation of concerns.
- Integrated Qdrant, Redis and MongoDB into AI services for vector search, semantic caching and persistent application data.
- Developed AI-powered automation workflows integrating LLMs with external APIs and tools to execute business operations from natural-language requests.
- Implemented intelligent workflows capable of identifying maintenance issues from user messages and automatically registering incidents in the company's management platform.
- Worked across backend implementation, API integration, application architecture, debugging and deployment throughout the development lifecycle.
- Collaborated with engineers on API contracts, technical decisions, code quality and integration of AI capabilities into production-oriented software.

Full Stack Developer

Jul. 2025 – Feb. 2026 *Pentabay Softwares*

Portugal

- Designed and developed full-stack applications for enterprise clients using JavaScript, React, Node.js, Express and MongoDB.
- Designed and implemented REST APIs for a Flight Ticket Management platform using a microservices architecture.
- Developed backend services and frontend components, taking ownership of features from implementation through API integration and testing.
- Worked with databases, API integrations and backend application logic in production-oriented environments.
- Used PM2 for process management, automatic restarts and application log monitoring in deployed environments.
- Collaborated with international Agile teams, participated in code reviews and contributed to technical documentation and software design decisions.

EDUCATION

Polytechnic Institute of Porto (ISEP)

Sep. 2023 – Jun. 2026

Porto, Portugal *B.Sc. Computer Engineering*

PROJECTS

Aegis – AI Agent Platform | *Python, FastAPI, LLMs, RAG, MCP, Qdrant, Redis*

- Designing an AI agent platform focused on integrating LLMs, retrieval, external tools and automated workflows into backend systems.
- Implementing AI agent capabilities including tool calling, planning, conversation memory and decision-making between answering, retrieving information and using external tools.
- Designing an agent orchestration architecture with specialized agents for retrieval, tool execution and response generation.
- Exploring Model Context Protocol (MCP) to expose external systems and operations as tools available to AI agents.
- Implementing RAG capabilities including semantic search, hybrid search, context selection and source attribution.
- Designing AI reliability and security features including prompt injection protection, authorization, tenant isolation, human approval workflows and AI evaluation.

AI Legal Assistant | *Python, FastAPI, RAG, LLMs, Qdrant, Redis, Docker*

- Designed and built an end-to-end AI application for answering domain-specific questions over internal documentation using LLMs and Retrieval-Augmented Generation.
- Implemented document ingestion, text processing, embedding generation, vector indexing and semantic retrieval pipelines.
- Designed the backend as modular services using FastAPI, Docker and clean architecture principles.
- Integrated Qdrant for vector search and Redis for caching and application performance optimization.
- Iterated on retrieval strategies, chunking approaches and prompts to improve response relevance and reduce unsupported model outputs.

Maritime Port Management System | *C, .NET, REST APIs, Frontend*

- Designed and developed a full-stack management system for a maritime shipping port, implementing both backend services and frontend functionality.
- Developed the backend using C and .NET, implementing business logic, REST APIs and core port management workflows.
- Designed and implemented the frontend to provide interfaces for managing port operations, entities and workflows.
- Applied object-oriented programming, software architecture and separation of concerns across the application.
- Implemented data persistence and database operations to support port-related information and business processes.
- Worked throughout the full development lifecycle, from requirements analysis and system design to implementation, integration, testing and debugging.

Alani E-Commerce Platform | *React, TypeScript, Node.js, Express, MongoDB, Stripe*

- Designed and developed a full-stack e-commerce platform from the ground up, covering frontend, backend, database and payment integration.
- Built REST APIs for product management, customizable products, shopping cart and order processing.
- Integrated Stripe payments and automated business workflows for order management.

TECHNICAL SKILLS

Languages : C#, Python, TypeScript, JavaScript, SQL, Java

.NET & Backend : .NET, ASP.NET Core, FastAPI, Node.js, Express, REST APIs, Microservices, Entity Framework Core

Software Engineering : Object-Oriented Programming, SOLID Principles, Clean Architecture, API Design, Software Architecture, Design Patterns, Testing, Git, Code Review, Agile Development

AI & LLMs : Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Vector Search, Prompt Engineering, Context Engineering, Tool Calling, Function Calling, AI Automation, MCP

Databases : PostgreSQL, MongoDB, Redis, Qdrant

Cloud & Infrastructure : AWS, Docker, CI/CD, Linux, PM2

Frontend : React, TypeScript, HTML, CSS

CERTIFICATIONS

Cisco Networking Academy

2025 *Introduction to Cybersecurity*

LANGUAGES

Portuguese — Native

English — Advanced